DGMS 2019 • SECTION H: ICHNOLOGY, REMOTE SENSING & GIS

Question 68: Vector GIS Spatial Analysis & Point-in-Polygon Overlay Operations

Geospatial Overlay Geometry, Topological Relationships, and Spatial Join Attribute Binding

OFFICIAL EXAMINATION QUESTION

In a GIS database, the location of meteorological station with respect to a specific land use can be identified using

(a) Line to polygon overlay

(b) Polygon on polygon overlay

(c) **Point in polygon overlay [CORRECT]**

(d) Line on line overlay

VERIFIED ANSWER KEY

Correct Option: (c) Point in polygon overlay — In vector GIS spatial modeling, a meteorological station is represented geometrically as a zero-dimensional **point feature** (x, y coordinate pair), while a land use zone (e.g., forest, agriculture, urban) is represented as a two-dimensional **polygon feature** (areal parcel). Determining which land-use parcel encompasses a meteorological station requires a **Point-in-Polygon overlay** (or spatial join).

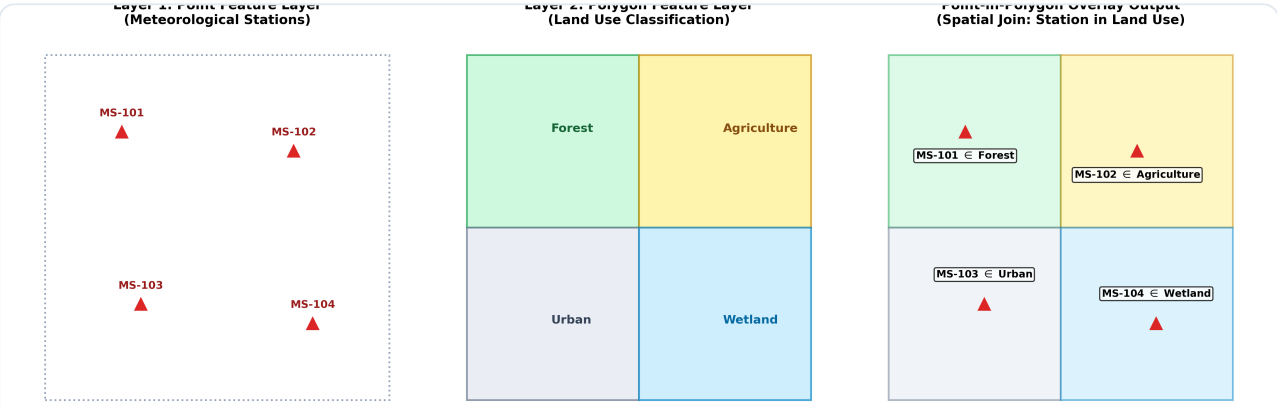


Figure 68.1 | Mechanics of Point-in-Polygon Vector Overlay in GIS: (Left) Point feature layer containing discrete weather monitoring stations. (Center) Polygon feature layer containing land use parcels. (Right) The resulting Point-in-Polygon overlay (spatial join) outputs a point dataset where each station has inherited the attribute properties (e.g., Land Use Class = Forest, Agriculture) of the surrounding polygon.

1. Vector Data Model Geometry & Spatial Overlay Principles

In a Geographic Information System (GIS), vector data structures represent real-world spatial phenomena using three fundamental topological primitives:

- **Points (0D):** Discrete locations with no length or area, defined by a single coordinate pair (x, y). Examples: Meteorological stations, groundwater boreholes, seismic observation stations, mineral deposit localities.
- **Lines** / **Polylines (1D):** Linear features with length but no width, defined by ordered sequences of vertices. Examples: Fault lines, river channels, transport networks, geological strike lines.
- **Polygons (2D):** Enclosed two-dimensional planar areas with perimeter and area, defined by closed boundary lines. Examples: Land use / land cover (LULC) classes, lithological units, watershed catchment basins, forest zones.

2. Types of Vector Spatial Overlay Operations

Vector overlay combines the geometries and attributes of two distinct thematic layers sharing the same spatial coordinate reference system (CRS):

Overlay Operation	Input Layer 1	Input Layer 2	Output Geometry	Typical Geological / Environmental Application
Point-in-Polygon	**Point (0D)** *e.g.,*	**Polygon (2D)**	**Point (0D)**	**Correct.** Identifies which polygon contains each point and appends the polygon attributes

Overlay Operation	Input Layer 1	Input Layer 2	Output Geometry	Typical Geological / Environmental Application
Overlay (Option c)	*Weather Station*	*e.g., Land Use*		(e.g., Land Use Type, Soil Type) to the point attribute table.
Line-in-Polygon Overlay (Option a)	Line (1D) *e.g., River, Road*	Polygon (2D) *e.g., Forest, Soil*	Line (1D)	Splits lines at polygon boundaries; determines length of road or stream segment flowing through different land use zones.
Polygon-on-Polygon Overlay (Option b)	Polygon (2D) *e.g., Lithology*	Polygon (2D) *e.g., Slope Class*	Polygon (2D)	Intersects or unions two areal layers (e.g., Union, Intersect, Identity) to create new polygon fragments combining both attributes for landslide hazard zonation.
Line-on-Line Overlay (Option d)	Line (1D) *e.g., Roads*	Line (1D) *e.g., Streams*	Point (0D) or Line (1D)	Identifies intersection points (bridge crossings) or overlapping line segments.

3. Point-in-Polygon Algorithms

Computational geometry engines in GIS software (e.g., QGIS, ArcGIS, PostGIS) evaluate point-in-polygon containment using rigorous mathematical algorithms:

- **Ray-Casting Algorithm:** A semi-infinite horizontal ray is projected from the test point (x_0, y_0) to infinity along the $+x$ axis. If the ray intersects the polygon boundary an *odd number of times*, the point lies **inside** the polygon. If it intersects an *even number of times* (or zero), the point lies **outside**.
- **Winding Number Algorithm:** Counts the total number of times the polygon boundary winds around the point; a non-zero winding number indicates the point is interior.

4. Comprehensive Option Analysis

- **(a) Line to polygon overlay:** *Incorrect.* Used for linear features (roads, streams, seismic profiles) intersecting polygons, not discrete point locations like meteorological stations.
- **(b) Polygon on polygon overlay:** *Incorrect.* Used when both input layers are planar polygons (e.g., overlaying a soil map on a geology map).

- **(c) Point in polygon overlay: Correct.** A meteorological station is fundamentally a point feature, and land use is a polygon feature. Point-in-polygon overlay directly identifies the specific land use parcel containing each meteorological station.
- **(d) Line on line overlay:** *Incorrect.* Used only to find intersections or overlaps between two linear networks.

5. Authoritative Textbook References

- **Kang-tsung Chang (2019)** *Introduction to Geographic Information Systems*, 9th Ed., McGraw-Hill Education, Chapter 11 ("Vector Data Analysis: Overlay Operations"), pp. 240–260.
- **Paul A. Longley et al. (2015)** *Geographic Information Science and Systems*, 4th Ed., Wiley, Chapter 13 ("Spatial Analysis"), Section 13.4 ("Spatial Overlays and Spatial Joins").
- **B. Bhatta (2021)** *Remote Sensing and GIS*, 3rd Ed., Oxford University Press, Chapter 12 ("Spatial Analysis and Modeling in GIS").
- **Burrough & McDonnell (2015)** *Principles of Geographical Information Systems*, 3rd Ed., Oxford University Press, Chapter 4 ("Vector Operations and Topological Overlays").